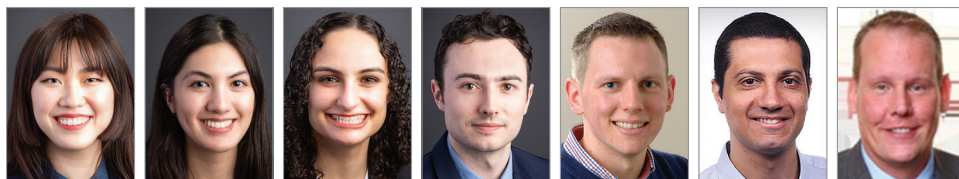


Catamenial Epilepsy

Understanding the underlying hormone pathophysiology and types of catamenial epilepsy is necessary for determining treatment options.

By Eunhae Yeo, BS; Emily Smith, BS; Natalie Ramsy, MS; Nicholas Peterman, BS; Aaron Anderson, PhD; Fadi Mikhail, MD; and Graham Huesmann, MD, PhD



Sex hormones influence brain development, memory, and neurologic conditions.¹⁻³ In catamenial epilepsy (CE), seizure patterns correlate with phases of the menstrual cycle, suggesting that sex hormones that drive menstruation also influence seizure susceptibility. Epilepsy is the fourth most common neurologic disorder in the US and 10% to 80% of menstruating people with epilepsy are thought to have some form of CE.⁴ Management of this highly prevalent seizure pattern should be familiar to any physician who cares for menstruating people with epilepsy in any capacity.

Despite the pervasive nature of CE, there is an overall paucity of research on the pathophysiology and pharmacologic management of CE.^{5,6} This review provides a practical guide to choosing pharmacotherapy for CE, considering the limited available evidence. To optimally manage and treat CE, a clinician must determine an affected individual's underlying hormone pathophysiology. We propose a framework for this clinical decision-making process by defining the 3 patterns of CE, reviewing the hormonal pathophysiology for each pattern, and summarizing pharmacotherapy treatment options.

CE Definition and Prevalence

CE is seen in reproductive-aged people who menstruate with any seizure syndrome. Historically, research groups used different definitions of CE. This resulted in wide variations in reported prevalence, from as little as 12.5% to as much as 78%.⁷⁻⁹ Definitions of CE were clarified in the late 1990s by plotting the percentage of persons with greater seizure frequency vs multiples of greater seizure frequency, which yielded 3 different patterns of CE (Figure 1).¹⁰ Such graphing also showed that in prior literature, the variance in the proportion of individuals affected (from ~10% to ~70%) was related to higher or lower cutoffs being used for the definition of CE. The calculated inflection points on the graphed patterns of CE showed approximately a twofold

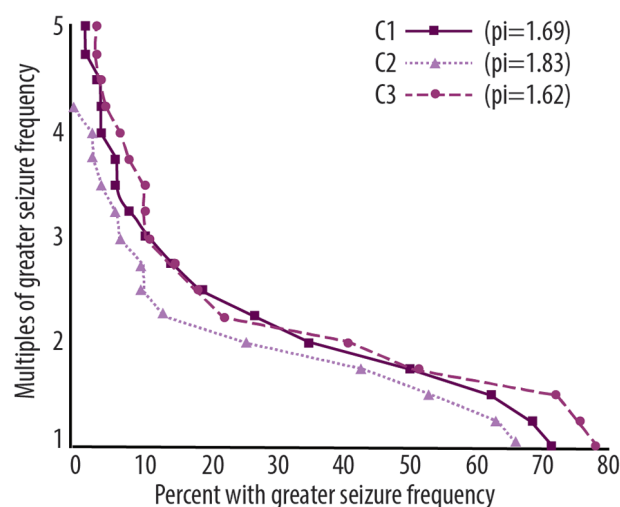


Figure 1. Frequency of Catamenial Epilepsy. The 3 patterns of catamenial epilepsy (C1, perimenstrual type; C2, periovulatory type; and C3, anovulatory type) were plotted as percent of people with higher seizure frequency vs multiples of greater seizure frequency. The point of inflection (pi) for each of the resulting S-shaped curves was calculated as the cut-off to optimally determine those with high menstrual seizure susceptibility from the overall population, all suggesting a twofold increase of seizure susceptibility for the designation of catamenial epilepsy.

increase in seizure frequency during the menstrual stage in question. This is the most widely accepted definition of CE in use today and used in this review.

CE Diagnosis and Patterns

Catamenial, from the Greek *katamenios* (monthly), describes a seizure pattern that correlates with the 4 phases of the menstrual cycle: menstrual, follicular, ovulatory, and

luteal (Figure 2).^{11,12} Clinical diagnosis starts with carefully reviewing the menstrual record and seizure histories and identifying a clear correlation between them. Estrogen is proconvulsive whereas progesterone is anticonvulsive.^{1,12} Therefore, phases of the menstrual cycle with increased estrogen-to-progesterone ratios cause seizure exacerbation. These phases include both the menstrual phase, in which progesterone is decreased, and the ovulatory phase, in which estrogen is increased. Menstrual phase seizure exacerbations are CE type 1 (C1), or the perimenstrual pattern. Ovulatory phase seizure exacerbations are CE type 2 (C2), or the

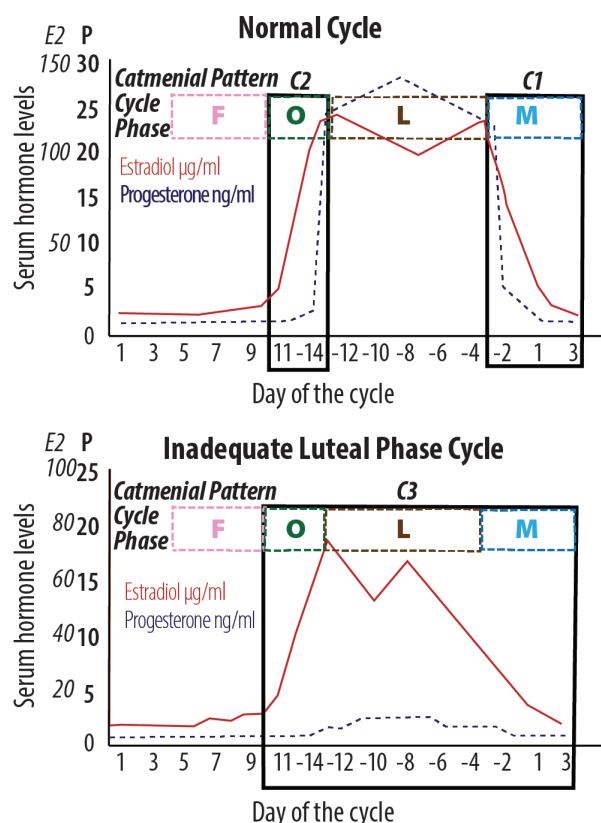


Figure 2: Menstrual Cycle and Catamenial Epilepsy Patterns. Serum hormone levels and menstrual cycle phases are displayed. Catamenial pattern subtypes (C1, perimenstrual type; C2, perioovulatory type; and C3, anovulatory type) are denoted at the cycle phase of seizure exacerbation. The luteal phase is numbered from days -14 to 0 to indicate its fixed 14-day distance from the day of menses, as opposed to the follicular period's variable length. Menses is defined as day 1. The menstrual stage (M) spans days -3 to 3 (first cycle day 25 to second cycle day 3), the follicular stage (F) spans days 4 to 9, the ovulatory stage (O) spans day 10 to day -13, and the luteal phase (L) spans Days -12 to day -4. Reproduced with permission from Zwog, AG. Frequency of catamenial seizure exacerbation in women with localization-related epilepsy. *Ann Neurol.* 2004;56(3):4.

perioovulatory pattern. Lastly, increased seizure frequencies during anovulatory cycles are designated CE type 3 (C3). Notably, the same individuals may have multiple types of CE.

Neuroactive Properties of Sex Hormones

Progesterone and estrogen mechanisms of action causing CE are complex and incompletely understood; as described, a simple understanding is that estrogen is proconvulsive and progesterone is anticonvulsive. Current understanding of the proconvulsive/anticonvulsive effects of estrogen and progesterone are described briefly here; more detailed reviews are available.^{1,13}

Progesterone is converted into metabolites called neurosteroids (eg, allopregnanalone and tetrahydrocortiosterone). These neurosteroids suppress seizures through unique interactions with GABA receptors (GABA-R) (Figure 3). Notably, altered GABA-R subunit populations have been implicated in several seizure models including CE. For instance, high progesterone levels are associated with high expressions of GABA-Rs with a δ subunit and decreased overall seizure susceptibility.¹¹

Estradiol-triggered convulsions in people with epilepsy who menstruate have been reported since 1959.^{3,14} Studies in rodents have even demonstrated cyclic epileptiform activity that correlates to fluctuating estrogen levels.¹ The exact mechanism of estrogen-triggered seizures remains unclear, however, because numerous studies also show that estrogen can suppress seizures.³ Some reasons for these paradoxical actions of estrogen include variable distribution of estrogen receptor subtypes in different brain regions and the broad effects of estrogen, which binds to glutamate, norpinephrine, and dopamine receptors, among others.¹⁵ Although neuroexcitatory properties of estrogen play a pathologic role in CE, estrogen excitatory activity is also involved in important

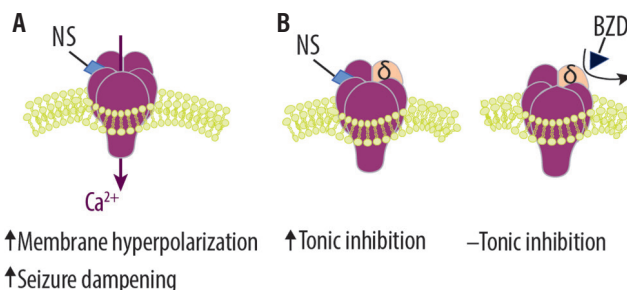


Figure 3: Progesterone and the GABA Receptor. Allosteric activation of GABA receptors by neurosteroids (NS) causes an influx of calcium ions resulting in membrane hyperpolarization and seizure dampening (A). GABA-receptors mediate 2 distinct types of inhibition (phasic and tonic), both of which contribute to overall seizure suppression. GABA receptors that contain a delta subunit (B) are particularly sensitive to activation by NS and resistant to benzodiazepines, which also suppress seizures through allosteric activation of GABA receptors.

cognitive processes (eg, memory, hippocampal synaptic plasticity, and memory-dependent task performance).¹⁶

A significant challenge in neuroendocrine research is that the relationship between serum neurosteroid levels and cerebrospinal fluid (CSF) neurosteroid levels is unclear. Reported CSF levels of neurosteroids vary between studies and may even be dependent on the brain compartment from which the CSF was sampled.¹⁷ It is also uncertain how much peripheral neurosteroids contribute to brain neurosteroid metabolism. Serum neurosteroid levels must therefore be carefully interpreted until this relationship is further clarified.

Pharmacotherapy for CE

There are no agents approved by the Food and Drug Administration (FDA) for CE, leaving pharmacologic management without clear guidelines. To guide the selection of CE pharmacotherapy, a comprehensive list of options with supporting evidence follows. Dosages, candidate CE subtypes, and other clinical details are given in the Table,¹⁸⁻²⁵ supported by the clinical experience of the authors. The body of literature on CE pharmacotherapy is small and many clinical trials are decades old. Many studies do not state their working definition of CE and, as such, application to the current understanding of CE may be limited. Nevertheless, these papers are the only source of data guiding CE therapy aside from anecdotal evidence, so it is important that this objective evidence is reviewed.

Nonhormonal Pharmacotherapy

Intermittent Benzodiazepines. Benzodiazepines, particularly clobazam, are frequently used to treat intractable epilepsies including CE. Clobazam is initiated 2 to 4 days before the expected menstruation phase in which seizure exacerbation occurs and continued for 1 days. This monthly dosing strategy is believed to help prevent benzodiazepine tolerance. Suitable treatment candidates are those with C1 and C2 subtypes. Individuals with C3 may not benefit because their irregular anovulatory cycles make treatment cycles challenging to plan.

Compared with other pharmacotherapeutic management options for CE, clobazam is relatively well-supported in the literature.⁴ In a placebo-controlled cross-over study using the described 10-day dosing regimen, 14 of 18 participants experienced seizure relief.¹⁸ A follow-up study confirmed that this treatment strategy had long-term efficacy, with 3 participants having seizure freedom for over 3 years.²⁶ Although this study was conducted before CE subtypes were established, the patterns described in the study match the C1 pattern. Notably, 2 participants experienced seizure exacerbation between benzodiazepine treatment windows, but still had an overall reduction in total seizure burden.

Acetazolamide. A carbonic anhydrase inhibitor, acetazolamide (AZM) has a long history of use in the treatment of

**TABLE. INVESTIGATIONAL TREATMENTS
OPTIONS FOR CATAMENIAL EPILEPSY**

Investigational treatments	Dosage/regimen	Candidate subtypes
Nonhormonal		
Clobazam	20-30 mg/day for 10-day monthly intervals, starting 2-4 days before menstrual phase associated with exacerbation	C1, C2
Acetazolamide	8-30 mg/kg/day in 1-4 divided doses, not to exceed 1g/day intermittently or continuously	C1, C2, C3
Hormonal		
Natural progesterone lozenge	50-200 mg 2 or 3 times/day on days 14-28 of menstrual cycle	C1
Ganaxolone	1500 mg/day	Indeterminate ^a
Clomiphene	15-100 mg/day, given on days 5-9 of menstrual cycle	C3
GnRH analogues	3-6 mg subcutaneous or intramuscular injection every 4 weeks ^b	C1, C2, C3
DMPA	150 mg intramuscular injection every 3 months	C1, C2, C3
Abbreviations: C1, perimenstrual type; C2, periovulatory type; C3, anovulatory type, DMPA, depot medroxyprogesterone acetate, GnRH, gonadotropin releasing hormone. ^a Insufficient studies in catamenial epilepsy to determine type for which agent may be effective ^b Consider daily progesterone for 2 to 3 weeks before first injection of gonadotropin-releasing hormone.		

refractory epilepsies.¹⁹ AZM is thought to increase accumulation of carbon dioxide in the brain, improving axonal stability.²⁷ All CE subtypes are potential candidates for treatment with AZM because it can be administered intermittently (during periods of seizure exacerbation) or continuously. The intermittent dosing strategy may be more beneficial for the C1 and C2 subtype, with continuous dosing more efficacious for C3. Adverse effects of AZM include its risk for tolerance, withdrawal seizures, and interactions with other medications.

Support for AZM use in CE in the published literature is relatively poor. In a retrospective analysis conducted in 2001, 8 of 20 people with perimenstrual (C1) reported significant decreases in seizure frequency, and 6 reported reduced seizure severity. No difference was seen between those who took AZM continuously vs intermittently. This study was based on

telephone questionnaires, making it subject to potential recall bias.²⁷ Other studies of AZM for CE are decades old and limited to only anecdotal evidence.¹⁹ More research is needed to conclusively determine the usefulness of AZM for CE.

Hormonal Pharmacotherapy

The choice of hormonal therapy is dependent on the regularity of an individual's menstrual cycles. For those with C1 and C2 with regular menstrual cycles, treatment of CE and maintenance of fertility can be achieved with progesterone and ganaxolone. For those with irregular menstrual cycles (C3), the treatment strategy can either be to induce menstrual regularity with clomiphene or to stop the cycle altogether with gonadotrophin-releasing hormone (GnRH) analogs or depot-medroxyprogesterone acetate (DMPA). For individuals who have regular menstrual cycles and do not wish to become pregnant, DMPA can also be used.

Natural Progesterone. Adjunctive natural progesterone lozenges can be considered for use in people with C1. Supplemental progesterone is thought to help prevent the perimenstrual progesterone withdrawal implicated in C1 seizure exacerbation. Frequent administration, however, is required due to the short half-life of progesterone. When considering treatment with natural progesterone, seizure exacerbations and menstrual cycles should be carefully documented to determine the appropriateness of progesterone therapy. Enzyme-inducing (EI) antiseizure medications (ASMs) may also increase progesterone metabolism and require modification of dosing. Side effects of progesterone include breakthrough bleeding and breast tenderness, and at high doses, there are also risks of CNS sedation, depression, and potentially breast cancer.²⁸

Multiple studies of progesterone treatment have shown varying levels of efficacy for CE.^{14,29-31} The most robust study of these was the first large-scale, randomized, placebo-controlled, double-blind clinical trial of cyclic adjunctive progesterone therapy for treatment of intractable seizures.²⁹ A total of 294 reproductive-age participants with ASM-resistant epilepsy were treated with progesterone (or placebo) lozenges along with their baseline ASM treatment. Although no significant difference was found between all who received progesterone and all who received placebo, a sub-group analysis of those with C1 showed a statistically significant response with progesterone lozenges vs placebo. For those with C1 exacerbations at least threefold in the perimenstrual vs other phases of menstruation, response to progesterone occurred in 37.1% vs 11.1% with placebo. These results should be interpreted cautiously because post-hoc subgroup analyses are subject to statistical bias. Nevertheless, because progesterone has a favorable safety profile, it may be worth trialing for individuals with severe and uncontrollable perimenstrual seizures.

Ganaxolone. A synthetic analog of allopregnanolone, ganaxolone is a neurosteroid metabolite of progesterone. After promising results in animal studies, ganaxolone has been investigated in several clinical trials as a potential adjunctive agent for intractable seizures.^{21,32} There is insufficient clinical information, however, to guide which CE subtypes are best suited to treatment with ganaxolone.

There are no published clinical trials of ganaxolone in people with CE. In a large, randomized, double-blind, placebo-controlled, phase 2 study of ganaxolone as adjunctive therapy for uncontrolled partial-onset seizures in adults, however, 68% of participants were identified as women, which the investigators attributed to "the perceived benefit [of ganaxolone] for women who have catamenial epilepsy."²¹

This study demonstrated that 1,500 mg/day ganaxolone resulted in a significant reduction in mean weekly seizure frequency compared with placebo. These results support ganaxolone as a potential safe and well-tolerated treatment option for CE, although more research is needed before a definitive recommendation can be made.

Clomiphene. The selective estrogen receptor modulator clomiphene is typically used to treat hypogonadism and has been used to regulate menstrual cycles.²² People with C3 are potential candidates for clomiphene citrate for seizure treatment, although there is little evidence to justify use in CE. There is a single study in the literature from 1987 that reported an 87% decrease in seizure frequency in 10 of 12 participants.²³ Because clomiphene is a pregnancy category X drug, pregnancy screening must occur throughout use. Long-term clomiphene use can also lead to increased risk of endometrial cancer.²²

GnRH Analogs. GnRH analogs (eg, goserelin and triptorelin) cause the suppression of pituitary gonadotropins and prevent monthly menstruation. Compared with oral contraceptives, which also stop menstruation, the GnRH analogue metabolism is unaffected by ASMs. Evidence for GnRH analogs for CE is strongest for the C1 subtype but considering that GnRH analogs work to stop menstrual cycles entirely, those with C2 and C3 may potentially benefit as well. When initiating GnRH analogs, there may be a transient spike in estrogen with subsequent seizure exacerbation before the pituitary gland starts down-regulating secretion. Daily progesterone for 2 to 3 weeks following the first injection of GnRH analogs is recommended to minimize this flare.⁴

Goserelin is relatively well supported for treatment of CE. A case report details a person whose hospital admissions for intractable CE status epilepticus were reduced from 10 to 3 visits/month after goserelin administration every 4 weeks.²⁴ Successful treatment of 10 individuals with 3.75 mg triptorelin every 4 weeks, for a mean 11.8 weeks has also been reported,²⁵ with 8 of 10 individuals experiencing improved seizure control. In this study, 3 individuals became almost seizure-free.

Follow-up of the same group showed that seizure reduction with GnRH treatment occurred in 11 out of 16 individuals with CE, further supporting the efficacy of this drug.³³

DMPA. A progesterone-only contraceptive, DMPA is injected intramuscularly every 3 months and reduces seizures when used in doses that produce amenorrhea. DMPA is therefore thought to reduce CE by suppressing the menstrual cycle. All catamenial subtypes potentially benefit from this treatment. In our clinical experience, we have observed an overall improvement in CE seizure control with DMPA. Individuals consistently experiencing catamenial seizures refractory to standard ASMs are referred to obstetrics and gynecology (OB/GYN) for DMPA administration, after confirming the individual does not desire pregnancy. Typically, a trial of synthetic progesterone pills is first given to monitor for unwanted side effects (eg, mood changes or cramps). If progesterone is well-tolerated, treatment advances to DMPA, and all of our patients who have had this treatment have had seizure reduction if not complete cessation of seizures with this treatment.

There are few published data on DMPA for CE. In a small study published in 1984, intramuscular medroxyprogesterone acetate was administered to 14 people with uncontrolled CE. Of 11 who became amenorrheic, 7 reported seizure reduction.³⁴ A more recent study surveyed 1,144 people taking ASMs about the effect of their contraception methods, including DMPA, on their seizures.³⁵ The impact of DMPA on seizures was mixed, with evidence that DMPA increased seizures in some but decreased them in others. Overall, more research is needed to conclusively determine the effectiveness of DMPA on CE.

Special Considerations for Hormonal Treatments. There are complex metabolic interactions between endogenous sex hormones, hormonal pharmacotherapy, and traditional ASMs. It is therefore critical to be vigilant for possible interactions as all 3 are important components of managing CE. For example, EIASMs can increase hepatic metabolism of estrogen and progesterone,³⁶ which may disrupt the balance of sex hormones and potentially exacerbate catamenial seizures. Concurrent administration of EIASMs with oral contraceptives can also decrease contraceptive bioavailability and efficacy. Clobazam, established as an effective treatment for CE, also has EI properties and therefore should be used carefully in people using oral contraceptives. Interestingly, EIASMs have not been found to affect the efficacy of DMPA injections, levonorgestrel-releasing intrauterine systems, or copper-containing intrauterine devices, making these potentially better contraceptive options for people with CE.³⁷ Oral contraceptives can also affect serum levels of ASMs. Concurrent administration of estrogen-containing oral contraceptives with lamotrigine has been found to increase lamotrigine metabolism and lower serum levels of lamotrig-

ine.³⁸ Individuals taking valproate with hormonal oral contraceptives have increased seizures compared with those taking valproate with nonhormonal oral contraceptives.²¹

Conclusion

Many of the clinical studies necessary to understand possible treatments and dosing strategies for CE were conducted before the establishment of CE pathophysiology. Because of this chronologic mismatch, the current body of evidence lacks consistent definitions, proper confounding variable control, and a cohesive model for CE. This review sought to compile these discordant studies and place them within the context of the current literature to support a merging of the C1, C2, and C3 patterns with differential treatment strategies originally proposed before these subtypes were established. This approach is intended to guide physicians in the utilization of current CE pathophysiology to form more evidence-based treatment strategies for people with CE patients. ■

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